Estimating the CCME Water Quality Index under Partial Measurement Availability: A Controlled Comparison of Random Forest and Gradient Boosting

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Abstract—Water quality indices condense several physical and chemical measurements into a single interpretable score through a fixed formula. This creates a problem when such an index becomes a supervised learning target: a model trained on the measurements that define it recovers the formula rather than learning anything about water quality. We demonstrate this on 2,818,006 harmonised surface-water records spanning five countries and eight decades, where gradient boosting reaches $R^2 = 0.9997$ against the CCME Water Quality Index – a figure that reflects function approximation, not prediction. We therefore reframe the task as estimating the index when measurements are unavailable, the case in which the formula cannot be evaluated and an estimate is genuinely required. Across four availability regimes, performance falls to $R^2 = 0.834$ when one measurement is withheld and 0.573 when four are, with mean absolute error rising from 0.100 to 8.596 index points. XGBoost leads Random Forest at every level on identical site-grouped test data, though on a larger training budget. We further identify an upstream median-imputation artefact affecting up to 31.6% of individual columns, concentrated almost entirely in one country and confounded with the target, and show by ablation that it does not warrant inclusion as a feature.

Index Terms—water quality index, CCME, gradient boosting, random forest, target leakage, missing measurements, tabular machine learning

I. Introduction

Routine environmental monitoring produces large volumes of physical and chemical measurements that must be reduced to an interpretable statement about water quality. Water quality indices serve this purpose by aggregating several parameters into a single dimensionless score, and the index published by the Canadian Council of Ministers of the Environment (CCME) is among the most widely adopted [1]. Such indices are attractive targets for supervised learning: they are continuous, widely reported, and accompanied by exactly the tabular measurement data on which tree ensembles perform well [2], [3].

That attraction conceals a methodological trap. A water quality index is not an observed quantity but a deterministic function of its inputs. When a model is trained on the same measurements from which the index was computed, it is not learning a relationship between environmental conditions and water quality; it is approximating a closed-form expression that is already known. Reported accuracy under this framing is high and the model is uninformative. In the taxonomy of Kaufman et al. [4] this is target leakage, and we found it in a published, widely reusable dataset: the distributed CCME values are reproduced from their eight contributing measurements with zero reconstruction error.

Our response is to change the question rather than abandon the data. When a monitoring programme fails to assay a parameter – because an instrument failed, a sample was lost, or the assay was never scheduled – the formula cannot be evaluated at all, yet an index value is still wanted. Estimating the index from the measurements that are available is a task with genuine content, immediate operational relevance, and no closed-form solution. We therefore retain the all-measurements setting only as an emulation benchmark and a check on the pipeline, and define the prediction problem over subsets of the measurement vector.

Working with a harmonised corpus of 2.8 million monitoring records from England, the USA, Ireland, China and Canada, we find the distinction is not academic. Withholding a single measurement moves gradient boosting from $R^2 = 0.9997$ to 0.834 and multiplies mean absolute error by a factor of 48. The step between those two numbers is the quantitative statement of how much apparent accuracy was formula reconstruction rather than inference.

Preparing the data surfaced a second issue of independent interest. The corpus reports no missing values, but

five of its eight measurement columns carry their own exact median as their most frequent value, at frequencies between 11% and 32%. The gaps were filled before publication, the fill is concentrated almost entirely in one country, and it is confounded with the target. We treat this as a finding to be measured rather than a defect to be silently inherited.

This paper contributes the following:

- A documented target-leakage pathway in a public water quality dataset, with its effect quantified by the gap between full and partial measurement availability.
- A partial-measurement evaluation protocol over four regimes, under a site-grouped split that admits no monitoring station to more than one partition.
- A controlled comparison of Random Forest and XGBoost at every availability level, with bootstrap confidence intervals and SHAP attribution.
- Detection and quantification of an upstream imputation artefact, together with an ablation showing that exposing it as a feature is not justified.

The remainder of the paper is organised as follows. Section II reviews water quality indices, tree ensembles and leakage. Section III describes the dataset, cleaning and exploratory findings. Section IV sets out the formulation, split and models. Section V reports results, Section VI states limitations, and Section VII concludes.

II. Related Work

A. Water Quality Indices

Composite indices reduce a vector of measurements to one interpretable number. The approach originates with Horton [5] and was formalised in the National Sanitation Foundation index of Brown et al. [6]; Abbasi and Abbasi [7] survey the family that followed. The CCME index [1] combines three factors – the proportion of variables failing their guideline (scope), the proportion of individual tests that fail (frequency), and the magnitude by which they fail (amplitude) – as a normalised distance from an ideal state. Lumb et al. [8] review its behaviour and note its sensitivity to the choice of guideline thresholds and to the temporal window over which frequency is computed, both of which matter for how the index is interpreted.

B. Machine Learning for Water Quality

Supervised learning has been applied to water resources along two distinct lines: predicting unmeasured or delayed constituents from sensor surrogates, and modelling composite indices directly for automated watershed scoring. Ahmed et al. [9] and Bui et al. [10] report strong accuracy for index prediction with tree ensembles and hybrid models. Our reading of this literature is that the second line requires care that is not always taken: when the index is a known function of the model's own inputs, high accuracy is the expected outcome and carries little information about the model.

We retain the continuous formulation rather than predicting the five CCME classes. Discretisation discards variance precisely where it matters, since a transition from 79.9 to 80.1 crosses a class boundary while a shift from 50 to 95 within a class is invisible, and it converts a naturally skewed continuous target into a severely imbalanced classification problem in which the worst-quality records are under 5% of the corpus.

C. Tree Ensembles on Tabular Data

Random Forest [2] reduces variance by averaging decorrelated trees grown on bootstrap samples with random feature subsets, and rarely overfits as the ensemble grows. Its impurity-based importances are, however, biased under correlated or high-cardinality predictors, a caution raised by Strobl et al. [11] and echoed for hydrological applications by Tyralis et al. [12]; we therefore report SHAP attributions [13] rather than impurity importances. Gradient boosting [14] instead fits trees sequentially to the loss gradient, reducing bias and capturing higher-order interactions, and XGBoost [3] adds regularisation, shrinkage, column subsampling and sparsity-aware splitting.

Comparative claims between these families are frequently confounded by unequal experimental conditions – an extensively tuned boosting model set against a default-configured forest, or inconsistent partitions between the two. We hold the split, the preprocessing and the evaluation contract identical across models and availability regimes, so that the comparison reflects the algorithms rather than the effort spent on them.

D. Leakage and Optimistic Validation

Kaufman et al. [4] distinguish leakage in training examples from leakage in features, where a predictor encodes information about the target that would not be legitimately available. A target computed by formula from its own predictors is the extreme case: the mapping is exactly recoverable and generalisation error approaches the numerical precision of the approximation. Roberts et al. [15] make the related argument that structured data demands structured validation, since random partitioning of spatially or temporally clustered records places near-duplicates on both sides of the split. Both concerns apply directly here, and shape the formulation in Section IV and the split in Section IV-B.

III. Dataset and Exploratory Analysis

A. Source and Scope

We use a harmonised surface-water monitoring dataset of 2,827,977 records across fourteen columns, covering England, the USA, Ireland, China and Canada [16], released under CC BY 4.0. Records span 2 January 1940 to 17 December 2023 over 61,689 monitoring sites. Each record carries eight measurements – ammonia,

biochemical oxygen demand (BOD), dissolved oxygen, orthophosphate, pH, temperature, nitrogen and nitrate – together with the CCME index value and its five-class categorical binning.

B. Cleaning

Cleaning screens rather than deletes. Observations falling outside the thresholds below are marked with boolean flags and retained, so that the decision to exclude them remains with the modelling pipeline and is visible in the results rather than buried in a data preparation step. We emphasise that these flags identify observations as potentially anomalous and warranting review; with the exception of the duplicate rule they do not establish that a reading is erroneous.

a) pH outside the conventional scale: 505 records (0.018%) fall outside $[0, 14]$, of which 504 lie above 14 (maximum 30.0) and one below 0. The 0–14 interval is the conventional scale for natural water [17], but it is an operational convention rather than a physical bound: pH is defined as $-\log_{10} a_{\mathrm{H}^+}$, and environmental waters as acidic as $\mathrm{pH} = -3.6$ have been measured in acid mine drainage [18]. These records are therefore flagged as implausible for routine ambient monitoring, not as impossible.

b) Temperature above 40°C: 7,284 records (0.258%) exceed 40°C, against an observed maximum of 98°C. This cutoff is a project-specific screening heuristic and not a published standard. Natural surface water does exceed it: the U.S. Geological Survey documents thermal springs discharging at 50°C and boiling at 93°C in a California creek [19]. The threshold is justified only by the character of this corpus, which is routine ambient monitoring rather than geothermal survey.

Two observations support treating these records as a recording artefact rather than as extreme measurements. First, 95.7% of them come from the USA, which constitutes 14.6% of the corpus; the flagged proportion is 1.68% of US records against 0.006% of English ones. Second, reading the flagged values as Fahrenheit places 99.99% of the US subset within an ambient 0–35°C band, shifting the median from 63.0°C to 17.2°C, which is consistent with the median of 11.1°C for unflagged US records. A unit-coding error confined largely to one national monitoring programme is therefore the most parsimonious explanation, though we note it remains a hypothesis that the distributed data alone cannot confirm. We retain the readings as supplied and correct nothing.

The flag is also insensitive to the exact cutoff. Candidate thresholds of 35, 40 and 45°C flag 7,576, 7,284 and 6,818 records respectively – a span of 758 records, or 0.027% of the corpus. No conclusion in this paper depends on the choice within that range.

c) Negative measurement values: 15 records carry a negative value in a column expected to be non-negative: orthophosphate (10), ammonia (2), biochemical oxygen demand (2) and pH (1). A concentration cannot be negative physically, but a negative reported result is not necessarily a transcription error, since blank-corrected analytical results near the detection limit can take small negative values. We were unable to establish the reporting conventions of the contributing monitoring programmes from the distributed data, so these records are flagged for review rather than attributed to a particular cause. We note that the screened column set includes pH, a logarithmic activity rather than a concentration; the single negative pH value is consequently captured by both this flag and the pH range flag.

d) Sub-zero temperature: 32 records fall below 0°C, with a minimum of −5.26°C. These are flagged separately and deliberately excluded from the aggregate anomaly indicator, because sub-zero water temperature is physically possible: water at 35 ppt salinity freezes near −2°C [20]. That is a statement about saline water in general and not about these particular records, 30 of which are English and 23 of which are rivers, with salinity unknown to us. The separation records our uncertainty rather than an assertion that the readings are valid.

e) Exact duplicates: 9,971 rows (0.35%) are identical across all 14 original columns and are removed, leaving 2,818,006 records. This is the only rule that deletes data. Duplicates are matched on the source columns alone, excluding the flags and the parsed date we derive, so that the definition depends on the distributed data rather than on our own processing. Identical records carry no additional information and would place the same observation on both sides of any partition.

f) No imputation at this stage: No values are imputed during cleaning. Imputation must be fitted on the training split alone, and that split does not exist at this point in the pipeline; performing it here would compute statistics across records destined for the test set.

C. An Upstream Imputation Artefact

The delivered file reports no missing values. Examining modal values shows this is an artefact: five of the eight measurement columns carry their own exact median as their single most frequent value – dissolved oxygen at 31.60% of rows, nitrate 25.33%, BOD 24.28%, pH 12.98% and temperature 11.02%. This is the signature of a median fill applied before publication. Only 39.11% of records are free of fill values.

Two consequences follow. First, the convention that imputation be fitted on training data alone was already violated before we received the data. Second, the fill is not uniformly distributed: England averages 2.04 filled measurements per record against at most 0.04 for every other country, while supplying 75.3% of the data (Fig. 1). Mean index value is 91.31 for fully measured records and 75.89 for records with three filled measurements, so fill status is confounded with the target. We therefore expose these as diagnostic flags and exclude them from

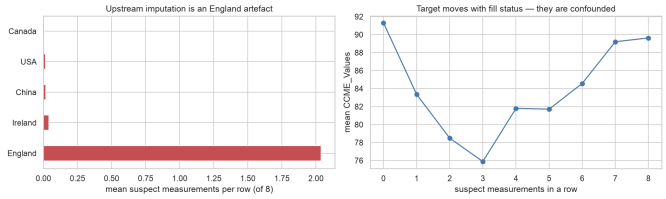


Fig. 1. Upstream median imputation is concentrated in one country (left) and is confounded with the target (right). Counts at six or more filled measurements are small and the apparent recovery there is noise.

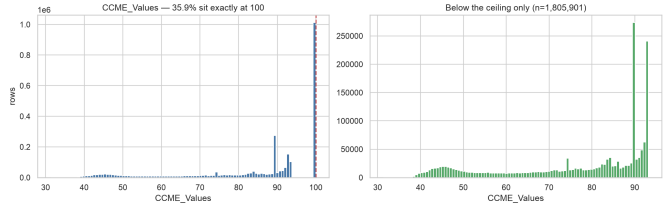


Fig. 2. The CCME index is bounded at 100, where 35.92% of records sit exactly (left); the distribution below the ceiling is shown at right.

the primary feature set, testing their effect by ablation (Table III).

D. A Bounded Target with a Mass at the Ceiling

The index is bounded above at 100 and 35.92% of records sit exactly at that bound; the 75th, 90th and 99th percentiles are all 100 (Fig. 2). The distribution has no upper tail, only a spike. The share at the ceiling varies from 20.9% (England) to 84.5% (USA), so the target's shape differs by country. Residuals above the ceiling are impossible by construction, making errors asymmetric.

We describe this as a bounded target with a point mass rather than a censored one, and the distinction matters for how it should be modelled. Censoring implies a latent value beyond the observation threshold that the recording process conceals. Here there is no latent value: by Equation (1), a record for which every measurement meets its guideline yields $F_1 = F_2 = F_3 = 0$ and therefore exactly 100. The ceiling is a property of the index's construction, not of its measurement, so Tobit-style censored regression is not the appropriate remedy. What the mass does imply is that the task mixes two questions – whether a record is fully compliant, and how far from compliance it falls otherwise – which a single pooled error statistic reports as one number.

E. Distributional Properties

The measurements are strongly right-skewed: ammonia reaches a skewness of 11.98 with kurtosis 222, BOD 10.21 and orthophosphate 8.94. A $\log(1+x)$ transform reduces ammonia skewness to 3.35. Tree ensembles are invariant to monotone transforms, so this matters chiefly for the Phase 2 neural model, but we evaluate the transformed features in ablation regardless.

Orthophosphate attains a Spearman correlation of -0.917 with the target, consistent with its tight guideline threshold dominating the index excursion term. Dissolved oxygen, by contrast, correlates almost not at all ($r = -0.031$). As dissolved oxygen is a "more-is-better" parameter, this suggests the upstream implementation treats its guideline as an upper bound; we return to this in Section VI.

Composition is uneven in two further respects. Effluent and sewage account for 22.1% of records – treatment-works outflow rather than natural surface water – with a mean index of 66.6 against 90.0 for rivers. And 82% of records fall between 2000 and 2019, with England's series beginning only in 2000.

IV. Methodology

A. Problem Formulation

Let $x \in \mathbb{R}^8$ denote the measurement vector – ammonia, BOD, dissolved oxygen, orthophosphate, pH, temperature, nitrogen and nitrate – and $y \in [0, 100]$ the CCME index. The index is defined [1] from a scope factor F_1, a frequency factor F_2 and an amplitude factor F_3 as

$$y = 100 - \frac{\sqrt{F_1^2 + F_2^2 + F_3^2}}{1.732}, \tag{1}$$

where $\sqrt{3} \approx 1.732$ normalises the maximum attainable distance to 100. Because each record in this corpus is an individual measurement rather than an aggregated time window, the upstream pipeline sets $F_2 \equiv F_1$; we note this adaptation because it departs from the protocol as published and affects how the index should be read.

Equation (1) is deterministic in x. Learning $f : x \mapsto y$ from the complete vector is therefore function approximation, and its error is bounded below only by the capacity of the approximator. We instead study

$$f_S : x_S \mapsto y, \qquad S \subset \{1, \ldots, 8\}, \tag{2}$$

where x_S is the measurement vector restricted to an available subset. For $|S| < 8$ the formula cannot be evaluated, no closed-form solution exists, and the model must exploit statistical structure among parameters, sites and water body types. We report $|S| \in \{8, 7, 6, 4\}$ and retain $|S| = 8$ as an emulation benchmark rather than a prediction result.

The subsets are chosen to be interpretable rather than exhaustive. The 7/8 regime withholds orthophosphate, which Section V shows to be the dominant driver of the index; 6/8 additionally withholds BOD, removing the heavy-organics signal; and 4/8 retains only the core physical-chemical suite of pH, dissolved oxygen, temperature and nitrate, approximating what a continuously deployed in-situ sensor package measures without laboratory assay.

B. Splitting

All experiments share one partition, generated once and persisted to disk. Every model at every availability level is validated and tested on byte-identical records, so reported differences between models are not attributable to differences in evaluation data. The training sets differ in size, and we state that explicitly in Section IV-D rather than claiming a parity we do not have.

Records are grouped by monitoring station (Area, 61,689 distinct sites) and whole groups are allocated to training, validation or test in a 70/15/15 ratio, with allocation stratified by country. No site appears in more than one partition. This follows directly from the concern raised in [15]: the same station sampled on consecutive dates yields near-identical chemistry, so a record-level random split would place near-duplicates on both sides and report a figure that reflects site memorisation rather than generalisation. Within each country, sites are shuffled under a fixed seed and assigned by greedy bin-packing in descending size order, each site going to whichever partition has the largest remaining deficit, which matches the target proportions as closely as integer site sizes permit.

One country resists this treatment. China contributes a single site (Hou Bay), which cannot be divided without breaking the zero-leakage guarantee, and is therefore allocated entirely to training. Chinese records consequently appear in the reported per-country breakdown as in-sample and are labelled as such; they are not evidence of generalisation.

C. Preprocessing and Features

The target is isolated before any transformation and never enters the design matrix. All fitted statistics – imputer medians, category encodings and scalers – are learned from the training partition alone and applied unchanged to validation and test. Masking for a partial-measurement regime removes the withheld columns and, importantly, any derived feature computed from them, so that no masked measurement re-enters through a ratio or a transform.

The raw feature set comprises the available measurements together with country and water body type. Site identifiers are excluded: with 61,689 levels under a site-grouped split, every test site is unseen by construction, so the variable can only act as noise. The engineered set adds logarithmic transforms of the skewed measurements – ammonia, BOD, nitrate, nitrogen and orthophosphate, whose skewness reaches 11.98 – and ratios between chemical parameters. Flags marking upstream-imputed values (Section III-C) are withheld from the primary feature set and evaluated separately by ablation, because they are confounded with country.

Programmatic checks assert, for every configuration, that the target does not appear among the features and that no group spans two partitions.

D. Models and Training Budgets

Three predictors are compared. A median predictor establishes the floor any model must clear. Random Forest is tuned over tree count, depth, minimum leaf size and features per split. XGBoost is tuned over learning rate, depth, subsample, column subsample, minimum child weight and L_1/L_2 regularisation, with early stopping on the validation partition.

The two ensembles are not trained on the same number of records. The partition assigns 1,986,397 records to training, and XGBoost uses all of them; Random Forest is fitted on a uniformly drawn subsample of 100,000, approximately 5% of the available training data. The reason is computational: Random Forest grows unpruned trees whose memory and fitting cost scale with the number of records retained at every node, and fitting the full partition was not feasible within our budget, whereas boosting's sequential, depth-limited trees are far cheaper per record.

This is a genuine asymmetry and we do not want it read as a like-for-like comparison. The two models see identical validation and test records, identical preprocessing and the same feature sets, but unequal training data. The reported margin in favour of XGBoost therefore reflects a combination of algorithmic difference and training budget, and part of the gap would likely close if Random Forest were given the full partition. We return to this in Section VI. Both expose the same training interface so that the Phase 2 neural model can enter the identical evaluation path. Hyperparameters are selected on validation only; the test partition is evaluated once, at the end.

E. Metrics

We report mean absolute error, root mean squared error and the coefficient of determination, computed through a single shared implementation so that every model is scored identically. MAE is reported alongside RMSE because the two respond differently to the heavy tails documented in Section III: RMSE is dominated by the small number of large excursions, while MAE describes the typical record. R^2 is reported for comparability with the water quality literature, with the caveat that its baseline is the target variance, which differs across availability regimes.

Confidence intervals are obtained by bootstrap resampling of the test partition, which makes the comparison between models a statement about sampling variability rather than a single point estimate. As noted in Section III-D, the target is bounded above and carries a large point mass at that bound; the metrics reported here are pooled across it, and Section VI states what that conflates.

V. Experiments and Results

A. Baseline and Full-Measurement Benchmark

Table I reports the three models under full measurement availability. The median baseline attains a test MAE

TABLE I
Phase 1 model comparison on the held-out test set, all-measurements regime; all figures are test-set. The baseline predicts the training median; Random Forest is fitted on a 100k subsample and XGBoost on the full training partition, both with engineered features.

Model	MAE	RMSE	R^2	Train (s)
Naive baseline	13.605	19.492	-0.0923	0.0
Random Forest	0.408	0.920	0.9976	1.7
XGBoost	0.100	0.328	0.9997	49.3

of 13.605 and $R^2 = -0.092$. The negative R^2 should not be read as evidence that the target is unusually hard: R^2 is defined against the mean predictor, which by construction sits at $R^2 \approx 0$, and a mean baseline on this data indeed gives -0.0006. The median scores worse on R^2 only because it minimises absolute rather than squared error, and correspondingly gives the lower MAE of the two (13.605 against 14.900). We use the median as the floor because MAE is our primary metric. Both ensembles exceed it by a wide margin, with XGBoost reaching $R^2 = 0.9997$ and a mean absolute error of 0.100 index points. Throughout, the two ensembles are trained on unequal budgets – Random Forest on a 100,000-record subsample, XGBoost on the full 1,986,397-record partition (Section IV-D) – so the margin between them is not a clean algorithmic comparison and should be read with that in mind.

This near-perfect score should be read as what it is. By Equation (1) the target is determined by the inputs, so the result measures how well a tree ensemble approximates a known piecewise function. We report it as a benchmark and as a check on the pipeline: a materially lower score here would indicate a defect upstream.

B. Partial Measurement Availability

Table II and Fig. 3 give the central result. Withholding a single measurement moves XGBoost from $R^2 = 0.9997$ to 0.834, and mean absolute error from 0.100 to 4.776 – a factor of 48. Withholding four leaves $R^2 = 0.573$ and an error of 8.596 index points, which spans roughly one CCME quality class. XGBoost leads Random Forest at every level, and the margin is stable at 2–4% of R^2 despite the twentyfold difference in training-set size, which suggests the ordering is not purely a budget effect even though its magnitude is not separable from one.

The gap between the 8/8 and 7/8 rows is the quantitative statement of the leakage problem. Almost all of the apparent accuracy in the full-measurement setting comes from the formula being reconstructible, not from any learned relationship that survives when an input is withdrawn.

C. Ablations

Table III isolates two design choices. Engineered features reduce XGBoost test MAE from 0.1517 to 0.1000, a 34% relative improvement, confirming that the logarithmic transforms and chemical ratios carry information beyond the raw columns. Adding the upstream-imputation flags changes test MAE only from 0.1517 to 0.1474. Given their confounding with country (Section III-C), this small gain does not justify their inclusion, and we exclude them from the primary configuration.

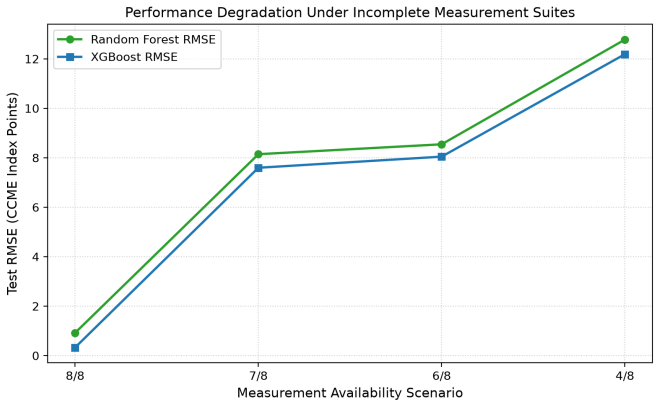


Fig. 3. Test performance against measurement availability. The step between 8/8 and 7/8 isolates the contribution of formula reconstruction.

D. Uncertainty and Error Decomposition

Table IV gives bootstrap confidence intervals. The intervals for Random Forest and XGBoost do not overlap on any metric, so the ordering between them is not an artefact of sampling.

Table V decomposes test error by country over all 415,804 test records. Performance is uniformly high in the full-measurement regime, but the sample sizes are extremely unequal: England supplies 76.8% of test records and Canada 0.14%, so the Canadian figures rest on 584 records and should not be read as a country-level comparison. England's share of the test set (76.8%) differs slightly from its share of the corpus (75.3%) because whole sites are allocated to partitions, so the split matches the target proportions only as closely as integer site sizes permit. China is absent: its single site is allocated wholly to training (Section IV-B), so it contributes no test records and cannot be evaluated out-of-sample.

E. Feature Attribution

Table VI ranks features by mean absolute SHAP value. Orthophosphate dominates at 8.813, nearly three times the next contributor, which matches both its Spearman correlation of -0.917 and its position in Equation (1): it carries the tightest guideline threshold and therefore the largest excursion term. The appearance of the engineered logarithmic transforms in the top five corroborates the ablation.

TABLE II
Degradation under partial measurement availability. The 8/8 row is a formula-emulation benchmark, not independent prediction; genuine estimation begins at 7/8. p is the feature count after masking; all figures are test-set.

Avail.	Model	p	MAE	RMSE	R^2
8/8	Random Forest	38	0.408	0.920	0.9976
8/8	XGBoost	23	0.100	0.328	0.9997
7/8	Random Forest	35	5.328	8.145	0.8093
7/8	XGBoost	20	4.776	7.597	0.8341
6/8	Random Forest	32	5.675	8.544	0.7901
6/8	XGBoost	17	5.209	8.045	0.8139
4/8	Random Forest	25	9.305	12.781	0.5304
4/8	XGBoost	10	8.596	12.190	0.5728

TABLE III
Ablation studies on XGBoost, all-measurements regime. p is the feature count; all figures are test-set.

Ablation	Config.	p	MAE	RMSE	R^2
Imputation flags	Excluded (primary)	10	0.1517	0.4005	0.99954
Imputation flags	Included	20	0.1474	0.3979	0.99954
Features	Raw	10	0.1517	0.4005	0.99954
Features	Raw + engineered	23	0.1000	0.3282	0.99969

TABLE IV
Bootstrap point estimates with 95% confidence intervals on the test set.

Model	MAE (Point [95% CI])	RMSE (Point [95% CI])	R2 (Point [95% CI])
Naive Baseline	13.6047 [13.5631, 13.6457]	19.4916 [19.4378, 19.5410]	-0.0923 [-0.0936, -0.0907]
Random Forest	0.4078 [0.4053, 0.4103]	0.9197 [0.9102, 0.9297]	0.9976 [0.9975, 0.9976]
XGBoost	0.1000 [0.0991, 0.1009]	0.3282 [0.3228, 0.3337]	0.9997 [0.9997, 0.9997]

TABLE V
Test error decomposed by country, over all 415,804 test records. China is absent by construction: its single site (Hou Bay) is allocated wholly to training, so it contributes no test records and cannot be evaluated out-of-sample.

Country	n	% of test	MAE	RMSE	R^2
England	319184	76.76	0.116	0.329	0.9997
USA	60787	14.62	0.062	0.394	0.9937
Ireland	35249	8.48	0.019	0.123	0.9996
Canada	584	0.14	0.176	0.561	0.9982

TABLE VI
Ten most influential features by mean absolute SHAP value (XGBoost, 8/8 regime).

Feature	Mean |SHAP|
Orthophosphate (mg/l)	8.813
log_Orthophosphate	3.062
Ammonia (mg/l)	2.455
Biochemical Oxygen Demand (mg/l)	1.061
log_Ammonia	0.691
Nitrogen (mg/l)	0.267
pH (ph units)	0.229
log_BOD	0.178
Temperature (cel)	0.160
N_P_ratio	0.110

VI. Limitations

a) Target leakage is mitigated, not eliminated: The 8/8 regime reconstructs the index formula and we report it as a benchmark rather than a prediction result. Even at 7/8 and 6/8, the retained measurements are formula contributors, so part of the remaining performance is partial formula reconstruction rather than environmental inference. A fully independent regime – predicting the index from metadata and spatiotemporal context alone, with no formula contributors – would bound this, and we did not run it.

b) A point mass at the ceiling is reported as one number: With 35.92% of records at the upper bound, pooled R^2 and RMSE conflate two tasks: identifying records that are fully guideline-compliant, and estimating how far the remainder fall short. The figures reported here are pooled. Metrics separated by ceiling status would characterise the models more honestly, particularly in the low-availability regimes where identifying compliance is likely the easier half of the problem. We note that this is a bounded target rather than a censored one (Section III-D): no latent value exists above 100, so the remedy is stratified reporting, not a censored-regression estimator.

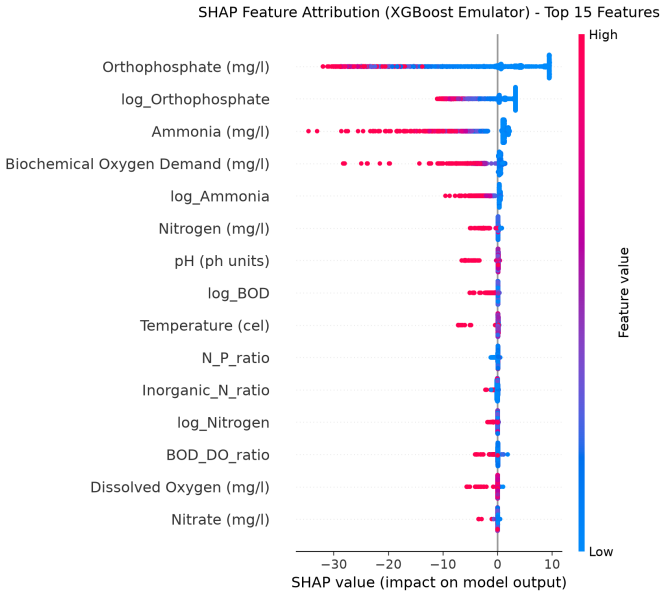


Fig. 4. SHAP summary for the full-measurement XGBoost model.

c) Cleaning thresholds are screens, not determinations: The anomaly flags described in Section III-B identify observations for review; they do not establish that a reading is wrong. The 40°C temperature cutoff in particular is a project-specific heuristic rather than a published standard, since natural surface water exceeds it [19], and the 0–14 pH interval is a convention rather than a physical bound [18]. The Fahrenheit unit-coding explanation we advance for the high-temperature records is consistent with their national and numerical distribution but cannot be confirmed from the distributed data, and we did not establish the reporting conventions behind the 15 negative values. A 0.26% subset of records is therefore flagged on evidence we can describe but not fully adjudicate.

d) The original missingness pattern cannot be recovered: What we observe directly is a repeated-value pattern: five of eight measurement columns carry their own exact median as their modal value, at 11–32% of records (Section III-C). That upstream median imputation produced this pattern is an inference from that evidence, not an established property of the upstream pipeline, and we make no claim to have recovered which individual values were originally absent. Because a filled value is indistinguishable from a measured one in the distributed data, the missingness mechanism cannot be classified as MCAR, MAR or MNAR. Because the fill is concentrated in England, fill status, country and target are mutually confounded, and no split design removes this.

e) A probable defect in the upstream index: Dissolved oxygen correlates with the target at $r = -0.031$. Dissolved oxygen is a "more-is-better" parameter and would normally be among the most informative inputs. The upstream implementation appears to treat its guideline as an upper bound to remain below, inverting its contribution. We report the index as published and flag this rather than silently correcting it.

f) Composition and coverage: Effluent and sewage comprise 22.1% of records, so the dataset is not purely natural surface water. England supplies 75.3% of records and its series begins in 2000, so the corpus is not geographically or temporally balanced, and aggregate metrics are dominated by one country.

g) What the split does and does not establish: Because whole monitoring stations are allocated to a single partition, every test record comes from a site the model has never seen, and the reported figures are therefore estimates of generalisation to new stations. They are not estimates of generalisation across time: the split is not chronological, so a test site's records may fall in the same period as the training data, and nothing here speaks to extrapolation into future years or to conditions outside the observed range. The Chinese subset is a further exception, being single-site and allocated entirely to training (Section IV-B); its per-country figures are in-sample.

h) Unequal training budgets: Random Forest is fitted on a 100,000-record subsample while XGBoost uses the full 1,986,397-record training partition (Section IV-D). Validation and test records, preprocessing and feature sets are identical, but the training data is not. The margin reported in favour of XGBoost therefore confounds algorithmic capability with training-set size, and should be read as an upper bound on the true algorithmic gap. Equalising the budgets is the first experiment we would add.

i) China cannot be evaluated out-of-sample: China contributes a single monitoring site, which cannot be divided across partitions without breaking the grouping guarantee and is therefore allocated wholly to training. It contributes no test records and is absent from Table V. Nothing in this paper evidences performance on Chinese waters.

j) Subset selection: The four availability regimes are interpretable rather than exhaustive. There are eight ways to withhold one measurement and seventy ways to withhold four; we report one informative choice at each level rather than averaging over all of them. The degradation curve should therefore be read as evidence that performance collapses once formula reconstruction is impossible, not as a precise estimate of expected loss under arbitrary missingness.

VII. Conclusion

We set out to predict the CCME Water Quality Index from physical and chemical measurements and found that the conventional framing of that task does not survive examination. The index is a deterministic function of the measurements that define it, so a model given all eight

recovers a formula rather than learning anything about water quality. Measured directly, this framing yields $R^2 = 0.9997$ and a mean absolute error of 0.100 index points – figures that look like a result and are not one. We report them only as an emulation benchmark and as a check that the pipeline is sound.

Reframing the problem as estimation under partial measurement availability restores its content. Withholding orthophosphate alone moves XGBoost to $R^2 = 0.834$ and multiplies error by a factor of 48; withholding half the measurements leaves $R^2 = 0.573$ and an error of 8.596 index points, roughly the width of one CCME quality class. The step between the full and partial regimes is the quantitative statement of how much apparent accuracy was reconstruction rather than inference. Within this formulation XGBoost leads Random Forest at every availability level, on identical test records and preprocessing, by a margin that bootstrap intervals confirm is not a sampling artefact – though the two were trained on unequal budgets, so that margin bounds rather than measures the algorithmic difference. Engineered logarithmic and ratio features reduce error by 34%, while flags marking upstream imputation, being confounded with country, do not earn inclusion.

Two findings extend beyond this dataset. The first is that a published, reusable corpus reporting no missing values had in fact been median-imputed before release, in a pattern detectable only from the modal structure of its columns and concentrated almost entirely in one contributing country. The second is that dissolved oxygen, which should be among the most informative parameters in any water quality assessment, carries almost no relationship to the index as distributed – consistent with its guideline being applied in the wrong direction upstream. Both were found by examining the data rather than the documentation, and both would have been invisible to a study that reported only predictive accuracy.

Phase 2 extends this work to a regularised multilayer perceptron trained on the same partitions and preprocessing, which allows the tree ensembles and the neural model to be compared without the confounds that make such comparisons unreliable in the literature. The skewness documented in Section III is expected to matter there in a way it does not for trees, since neural models are not invariant to monotone transforms, and the engineered features will be ablated accordingly. The point mass at the target's ceiling also deserves direct treatment, through metrics separated by compliance status and potentially through a model that represents the boundary explicitly rather than regressing through it.

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